

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. **(L2)**

Assessment Tool: Code along exercise: Make your first HTML page together with CSS

Weightage: 10%

QUESTIONS: Total Marks: 10

Scenario: Web Development Standards Implementation

A start up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.

The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

A. `<media src="video.mp4"></media>`

- B. `<video src="video.mp4" controls></video>`
- C. `<movie src="video.mp4"></movie>`
- D. `<vid src="video.mp4"></vid>`

Snippet 2: Form Submission Methods

- A. GET
- B. POST
- C. PUT
- D. DELETE

Snippet 3: Navigation Elements

- A. `<div>`
- B. `<nav>`
- C. `<section>`
- D. `<header>`

Snippet 4: XML Syntax

- A. `<note><to>User</to><from>Admin</from>`
- B. `<note><to>User</to><from>Admin</from></note>`
- C. `<note><to>User</from></note>`
- D. `<note to="User"></note>`

Snippet 5: Web Communication Protocols

- A. FTP
- B. HTTP
- C. SMTP
- D. SNMP

Questions:

1. Select the **correct HTML tag for video embedding** and justify your answer.
2. Identify the **appropriate HTTP method for form submission** and explain why.
3. Choose the **proper semantic element for navigation** and justify its usage.
4. Identify the **correct XML syntax** and explain the error in other options.
5. Choose the **correct protocol for web communication** and justify your choice.

Code of Conduct:

*I, **Aaron Samuel S(192411022)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding ; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

1. Select the correct HTML tag for video embedding and justify your answer.

Answer: B. `<video src="video.mp4" controls></video>`

Justification:

- The `<video>` tag is the standard HTML5 element used to embed videos in a web page.
 - The `controls` attribute provides built-in playback controls such as Play, Pause, Volume, and Full Screen.
 - The other tags (`<media>`, `<movie>`, `<vid>`) are **not valid HTML elements**.
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2. Identify the appropriate HTTP method for form submission and explain why.

Answer: B. POST

Justification:

- The **POST** method is used to securely send form data to the server.
 - Data is sent in the request body rather than being appended to the URL, making it more suitable for sensitive information like passwords and personal details.
 - **GET** exposes data in the URL, while **PUT** is mainly for updating resources and **DELETE** is used for deleting resources.
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3. Choose the proper semantic element for navigation and justify its usage.

Answer: B. `<nav>`

Justification:

- The `<nav>` element is the semantic HTML tag specifically designed for navigation menus.
 - It improves website accessibility and helps search engines understand that the content contains navigation links.
 - `<div>` has no semantic meaning, `<section>` represents a thematic section, and `<header>` is used for introductory content.
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4. Identify the correct XML syntax and explain the error in other options.

Answer: B. `<note><to>User</to><from>Admin</from></note>`

Justification:

This XML is well-formed because:

- All opening tags have matching closing tags.
- Tags are properly nested.
- The root element (`<note>`) is correctly closed.

Errors in other options:

- **A:** Missing closing `</note>` tag.
 - **C:** Mismatched closing tag (`</from>` instead of `</to>`).
 - **D:** Incorrect closing tag (`</note` is missing the closing `>`).
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5. Choose the correct protocol for web communication and justify your choice.

Answer: B. HTTP

Justification:

- **HTTP (HyperText Transfer Protocol)** is the standard protocol used for communication between web browsers and web servers.
- It enables the transfer of web pages, images, videos, and other web resources.
- **FTP** is used for file transfer, **SMTP** for sending emails, and **SNMP** for network management.